Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology
B.Tech. | Sem-6 | Summer-2024

Course Code	:	2101CS621	**Date**	:	18-04-2024
Course Name	:	Machine Learning	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Classify Supervised and Unsupervised Learning.	**4**
	(B)	Define Machine Learning and explain its types.	**3**
		OR	
		Explain parameters and output for Train_Test_Split()	
	(C)	Define function approximation. Why to estimate f? How to estimate f?	**7**
		OR	
		Using the concept of Underfitting and Overfitting explain how we get U shape in Test Error.	
Q.2	**(A)**	Explain the steps involved in Principal Component Analysis.	**4**
	(B)	Explain any two metrics used to evaluate Linear Regression.	**3**
		OR	
		Explain any three properties of Linear Regression.	
	(C)	Explain linear regression and write a python code to implement linear regression (assume Sample Data)	**7**
		OR	
		Explain logistic regression write a python code to implement logistic regression (assume Sample Data)	
Q.3	**(A)**	Draw and explain the four cases of AUC and ROC graph	**4**
	(B)	Explain K-Nearest Neighbour Classifier.	**3**
		OR	
		Explain Naïve Bayes Classifier.	
	(C)	Explain the concept of Decision Trees with example.	**7**
		OR	
		Explain the different components of Confusion Matrix with example.	

Q.4	**(A)**	Classify Hierarchical and Non-Hierarchical Clustering.	**4**
	(B)	Explain support vectors in SVM.	**3**
		OR	
		What are outliers? Explain how DBSCAN algorithm is used for outlier Detection.	
	(C)	Explain K-Means clustering and write a python program to implement K-Means clustering algorithm on sample data.	**7**
		OR	
		What are the different linkages in dendrogram? Write a python program to implement a dendrogram on sample data.	
Q.5	**(A)**	Classify Feedforward and Back propagation in ANN.	**4**
	(B)	Explain activation Functions in ANN.	**3**
		OR	
		Classify between Shallow and Deep Neural Network.	
	(C)	Explain in detail how does the Neural Network get trained.	**7**
		OR	
		Explain with a diagram Neural Networks and its different layers.	

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